Geometric shape representations in human adults have syntactic structure

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Background

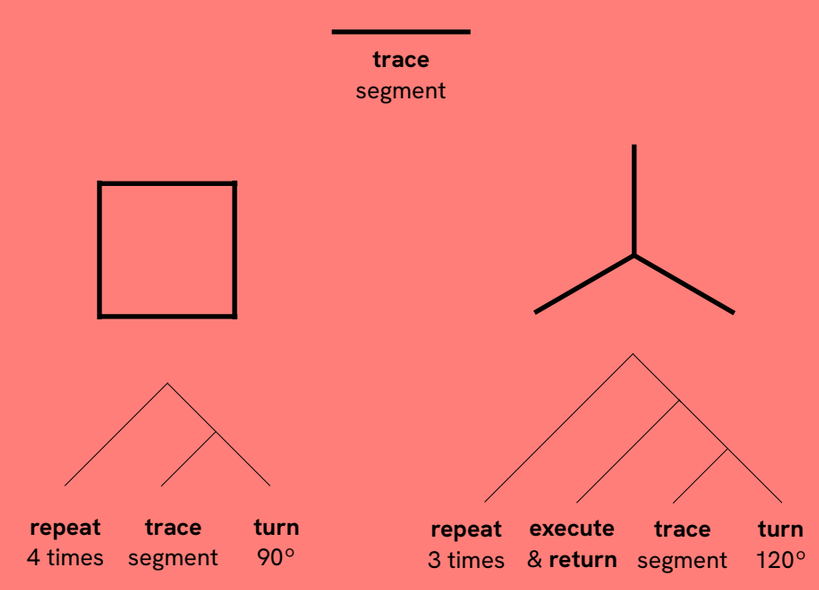
We build on the recent proposal that **geometric shapes are represented in a language of thought (LOT)** consisting of a handful of primitives that can be combined to recreate the encoded shape (Sablé-Meyer et al. 2022)

Adults' reaction times and error rates in encoding shapes were predicted by the **minimum description length (MDL)** of the hypothesized LOT

This fuses two aspects of geometric shape representations that are partly independent, **format** (LOT) and **selection** (MDL), and uses MDL data to argue for LOT—highly indirect evidence

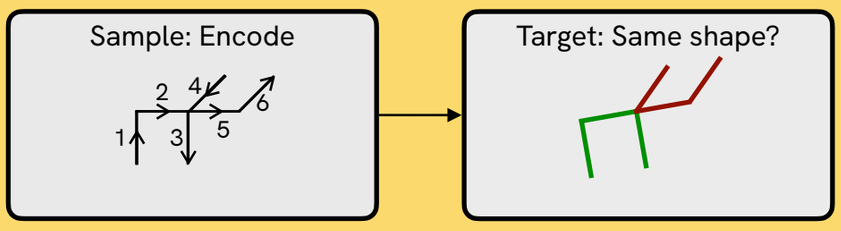
Current study provides **more direct evidence for tree structure** in geometric shape representations in 3 online experiments with adults

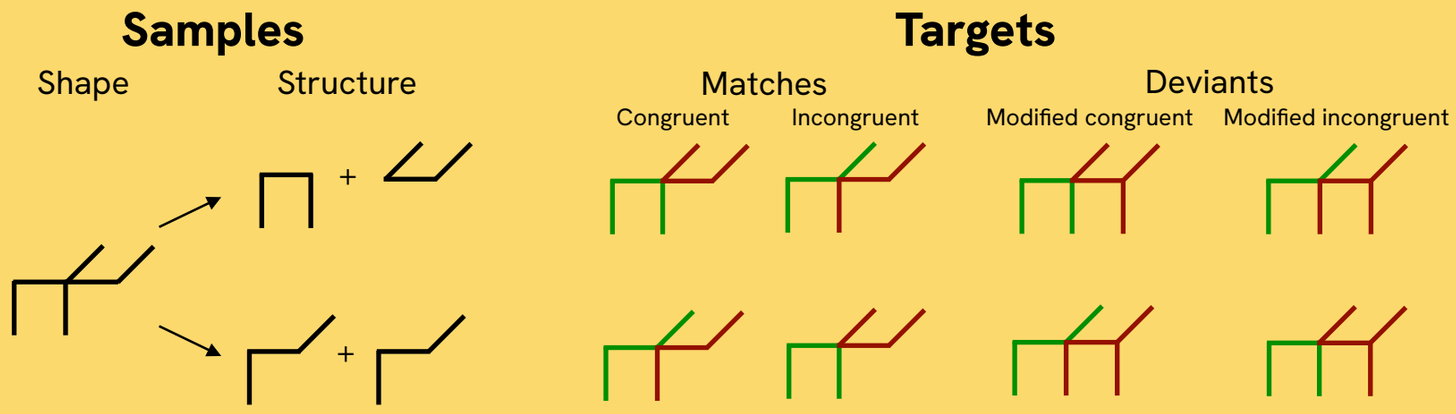
Hypothesis: Geometric shapes are encoded in a dedicated, syntactically structured language of thought



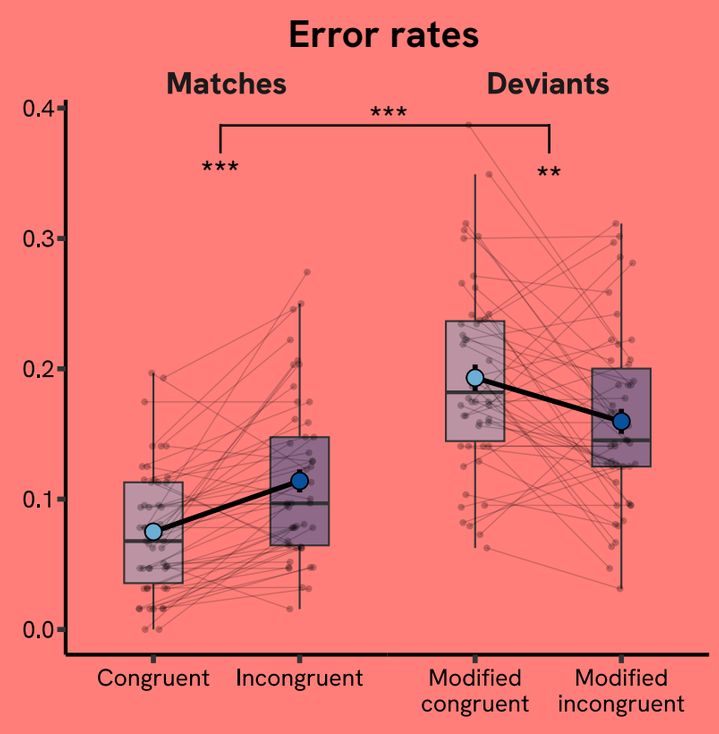
Experiment 1: Structural Ambiguity

50 subjects | 256 trials



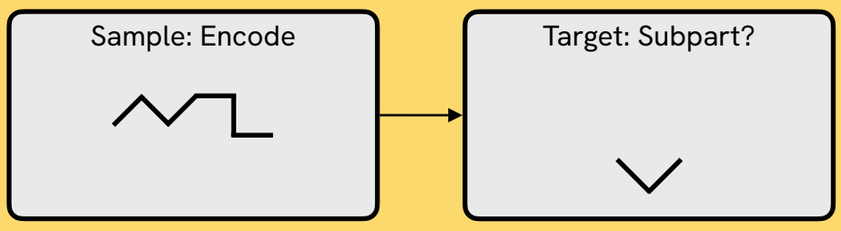


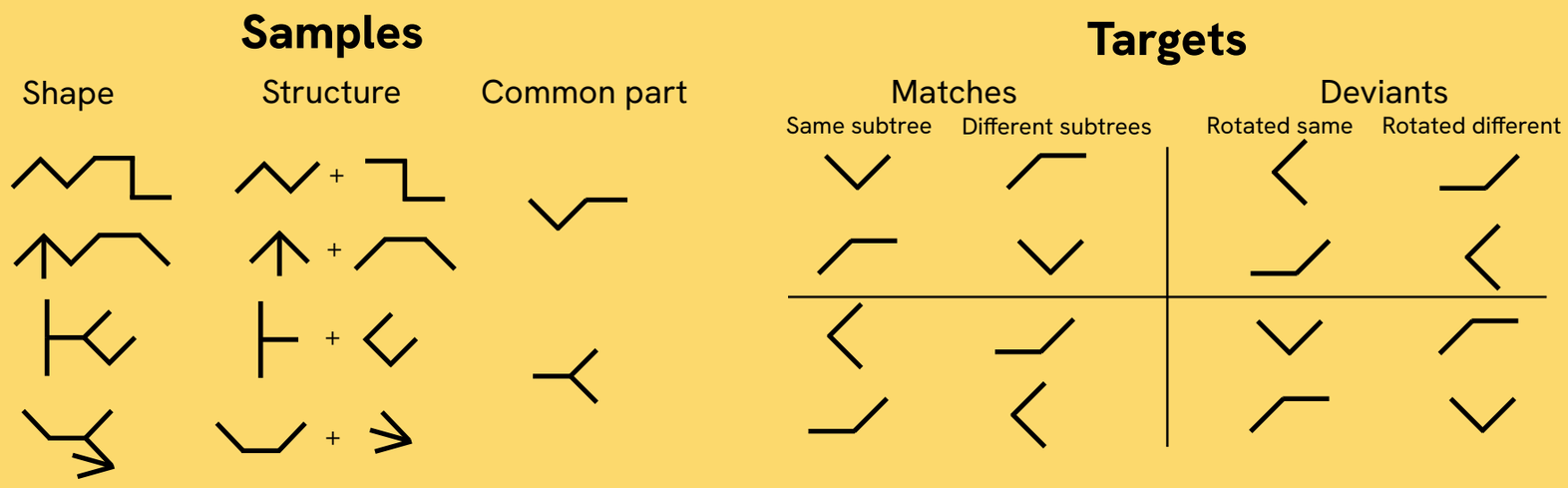
Different structured representations can be induced for the same shape



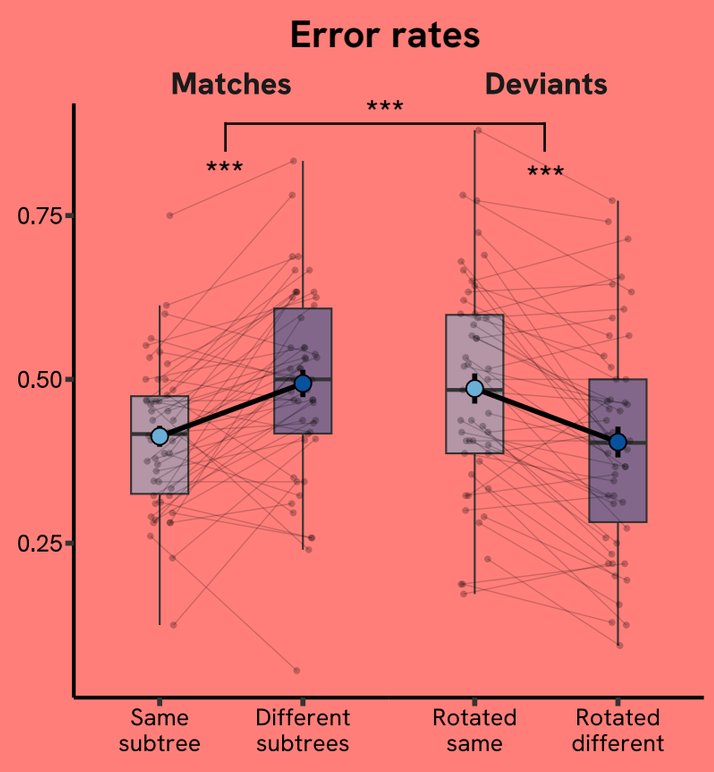
Experiment 2: Subtree Facilitation

50 subjects | 128 trials



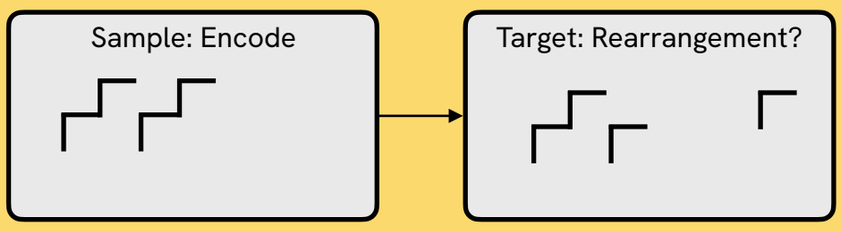


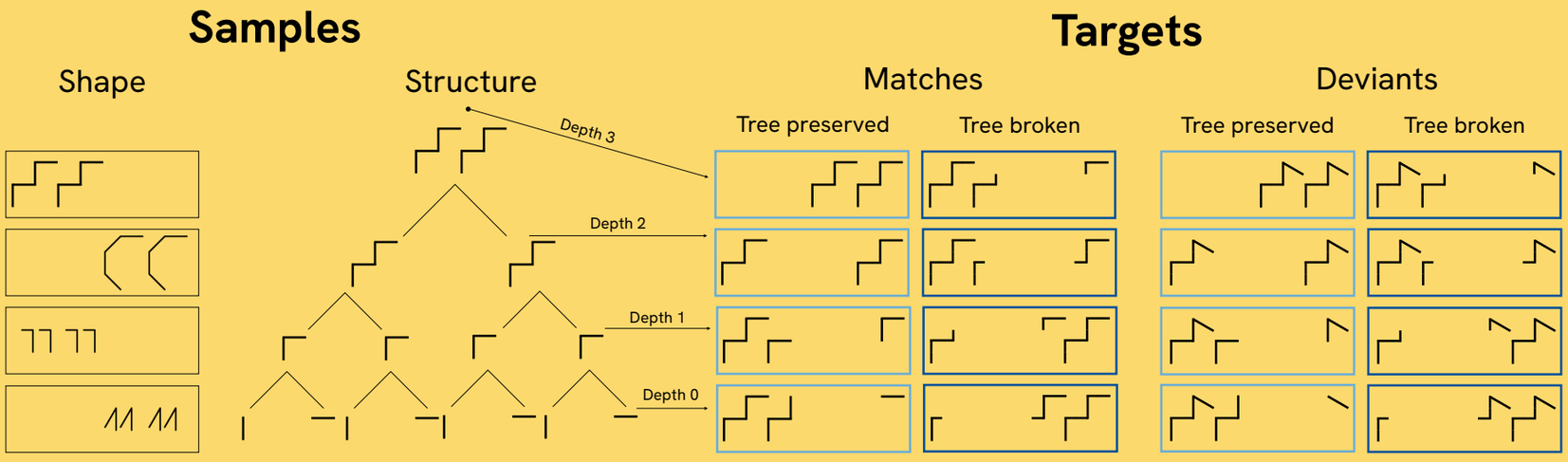
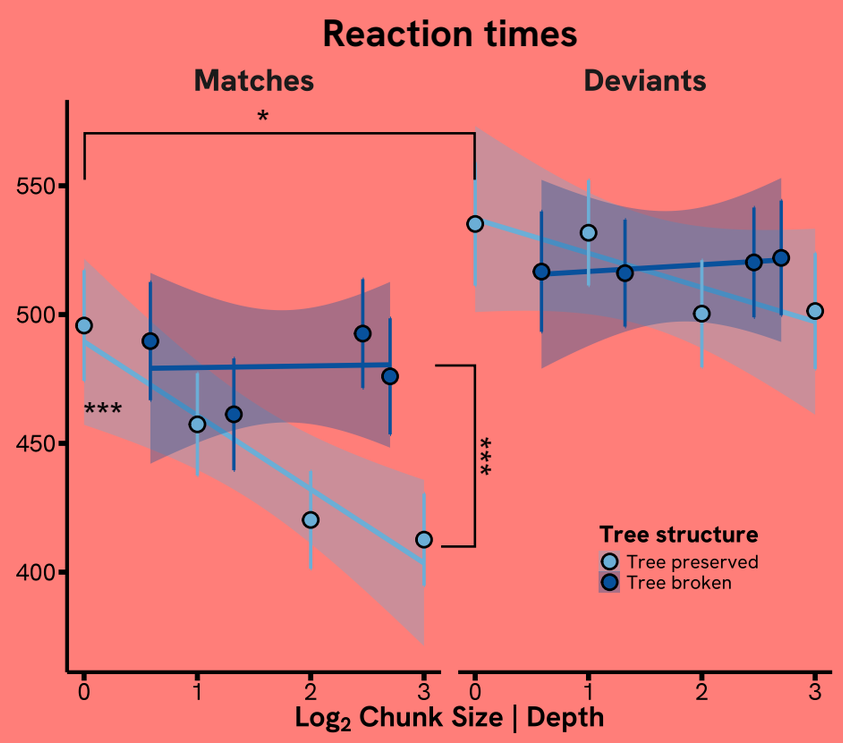
Subparts are easier to recognize when they belong to the same subtree



Experiment 3: Movement Depth

36 subjects | 256 trials





Shapes are easier to reconfigure when they are split higher up in the tree